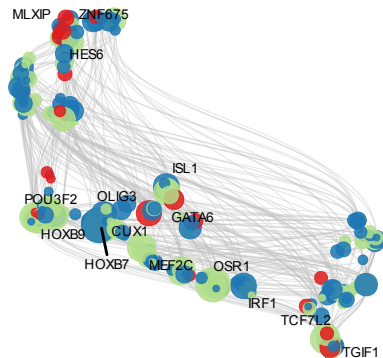


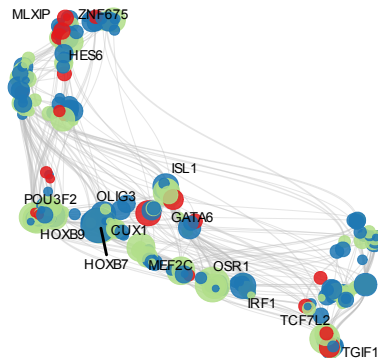
TF UMAP ... direct vs. indirect regulatory edges

Node colour: outgoing-edge dominance | Direct-dominant $\text{frac_direct} > 2/3$ | Indirect-dominant $\text{frac_direct} < 1/3$ | Labels: top 5 TFs per category by reverse PageRank Mixed

Direct edges only (TF ... promoter(TF))



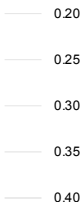
Indirect edges only (TF ... enhancer ... promoter(TF))



Reverse
PageRank



value



value